# Alhassan Alfarran

## Python / Full-Stack Developer (FastAPI, React + TypeScript)

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**Open to offers · on-site, hybrid or remote · full-time or contract**

### PROFESSIONAL SUMMARY

Building software commercially since 2024. In 2026 I built semantic and full-text search over technical documentation (FastAPI, sentence-transformers, FAISS, a custom inverted index): an archive of ~4.1k files (~1.5k documents with text), 0.18 s p95 response time, 739 automated tests, accepted by the client. In 2025 I was the lead developer of an internal web system for a network of 9 vehicle depots (React + TypeScript, Node.js/Express, MongoDB, Docker Compose). I develop with AI agents (Claude Code, Cursor, Codex): I set the task, review the changes, check them with tests and on real data, and deploy them.

### TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript

**Backend:** FastAPI, Node.js / Express, REST APIs, JWT authentication, background jobs, file uploads (Multer)

**Search & documents:** sentence-transformers, FAISS, inverted index, hybrid search; pypdf, python-docx, openpyxl; OCR (Tesseract, Poppler)

**LLM (prototypes built with AI agents):** RAG (FAISS + BM25, RRF), OpenAI-compatible APIs, MCP server

**Frontend:** React 18, TypeScript, Vite, Next.js, Tailwind CSS, responsive UI, framework-free HTML/CSS/JS

**Databases:** MongoDB (Mongoose); SQLite / PostgreSQL — university project (built with Codex)

**Infrastructure & quality:** Linux (Ubuntu, systemd), Docker / Docker Compose (basic), GitHub Actions, Git, pytest

**Data:** pandas, NumPy, scikit-learn — university projects

**Ledgerline (built with Claude Code):** Kotlin, Spring Boot, Kafka, PostgreSQL (event sourcing), ClickHouse, gRPC, OpenTelemetry / Tempo, Vue 3

**Depotwin (built with Claude Code):** Go, NATS JetStream, TimescaleDB, three.js / MapLibre, Expo / React Native, LightGBM, Prometheus / Grafana, k6

**Clearhire (built with Claude Code):** Angular, NestJS, PostgreSQL + pgvector, Redis, Solidity / Foundry, Kubernetes / Helm

### PROFESSIONAL EXPERIENCE

**Python Developer** — Elektroservis (Электросервис) Apr 2026 — Oct 2026

Contractor (civil-law contracts) · remote

- Built a search service for an engineering-documentation (PTO) archive: an engineer enters a cable mark, document code, equipment designation or a phrase and gets the document, the page, a text fragment and the network path.
- Hybrid search with no LLM generation: a custom inverted index plus FAISS semantic search on sentence-transformers embeddings; FastAPI backend, plain HTML/JS UI. PDF, DOCX, XLSX and RTF indexing (~4.1k files, ~1.5k with text); OCR of selected scanned pages (Tesseract + Poppler).
- Speed: code and mark queries from 9–10 s to 1.2–1.7 s, phrase queries from 55–70 s to 2–4 s; **0.18 s p95** after moving to the inverted index.
- Quality: **739** pytest tests and a 104-query control set (top-1/top-5); systemd service on Ubuntu Server with atomic index swap and rollback. Accepted by the client in September 2026.
- Built with AI agents (Codex, Cursor, Claude Code): gathered requirements, wrote the agent briefs, reviewed the changes and verified them on the production index.

**Full-Stack Developer (lead developer on the project)** — Avenue Group May 2025 — Dec 2025
Moscow, Russia · team of 3

- Designed the architecture and API of OKKP, an internal quality-control web system for 9 vehicle depots, and wrote most of the code (~177 of 183 commits in 2025).
- Violations and fines module with server-side filtering and pagination; modules for employees, driver call-backs, a document archive (Cloudinary with local-disk fallback) and notifications.
- Per-depot KPI tiles and in-browser Excel export (SheetJS); JWT authentication (bcrypt); responsive Tailwind CSS UI.
- Deployment with Docker Compose (MongoDB + API) on a Linux server; CI on GitHub Actions. Stack: React 18, TypeScript, Vite, Node.js, Express 5, MongoDB.

**Full-Stack Web Developer** — Junzi Tech Solutions Oct 2024 — Jul 2025

- Full-stack features for web applications with Node.js, Express, MongoDB and React.
- Cut the response time of a critical API endpoint by **~60%** (caching and a redesigned database query).

**Freelance Developer** 2024 — Present
Remote

- React / Next.js / TypeScript and Tailwind CSS web applications for clients.

## PROJECTS

**Ledgerline** — open-source platform, built with Claude Code Nov 2026
github.com/kyan9400/ledgerline · live demo: ledgerline.149-36-1-90.sslip.io

- Payments platform built like a bank's: a double-entry ledger on an event log (Kotlin, Spring Boot, PostgreSQL), card payments and payouts as sagas, an outbox into Kafka. Nothing lost or done twice over 11 chaos scenarios (40,461 requests); a LightGBM fraud model caught 88% (rules: 37%); ClickHouse analytics, a gRPC API with signed webhooks, OpenTelemetry tracing and alerts proven by fault injection.

**Depotwin** — open-source platform, built with Claude Code Nov 2026
github.com/kyan9400/depotwin · live demo: depotwin.149-36-1-90.sslip.io

- Live 3D digital twin of 240 simulated vehicles in Moscow: Go simulator, NATS JetStream, TimescaleDB, FastAPI WebSockets, React/three.js. LightGBM warns of overheating 4.5 days ahead (rules: 3.0); a local-LLM copilot over read-only queries (17/18 held-out questions); an offline-first Expo driver app; Prometheus, Grafana, k6 (200 req/s, 0 errors).

**Clearhire** — open-source platform, built with Claude Code Nov 2026
github.com/kyan9400/clearhire · live demo: 149-36-1-90.sslip.io

- AI recruiting: checks CVs (PDF, Word, OCR scans) against each requirement with the CV line as evidence, blind screening, a local LLM that must cite the CV, audit log anchored in Solidity. FastAPI, pgvector, Angular, Next.js, NestJS, Helm/Kubernetes. nDCG@10 0.97 on 312 synthetic CVs.

**WhyDoc** — prototype, built with Claude Code Oct 2026

- RAG over documents: answers cite the PDF page or Excel cell, and claims are checked against the source. FAISS + BM25 (RRF), FastAPI, LLM via an OpenAI-compatible API.

**Lumen Search** — open-source prototype, built with AI agents Oct 2026
github.com/kyan9400/lumen-search · live demo: kyan9400.github.io/lumen-search

- Hybrid search with citations over PDF, Word, Excel and scanned pages in English and Russian: BM25 + multilingual embeddings in FAISS, Tesseract OCR, FastAPI.

**Pathwise** — master's thesis, built with Codex 2026

- Recommender system for study programmes: FastAPI, scikit-learn (TF-IDF + cosine similarity), pandas, PostgreSQL or SQLite; Next.js front end.

## EDUCATION

**Master's Degree** — National University of Science and Technology MISIS, Moscow 2024 — 2026
Computer and Information Sciences

**Bachelor's Degree** — Ural Federal University, Yekaterinburg 2020 — 2024
Computer Science

## CERTIFICATIONS (SELECTED)

- Full-Stack Web Development with Angular — Coursera, HKUST
- Server-side Development with Node.js, Express and MongoDB — Coursera
- Introduction to SQL; Understanding and Visualizing Data with Python — Coursera, University of Michigan
- Machine Learning for All — Coursera, University of London
- Programming with Google Go Specialization — Coursera
- JavaScript Algorithms and Data Structures; Data Visualization — freeCodeCamp
- ICDL — International Computer Driving Licence

## LANGUAGES

**Arabic** — native · **English** — C1 (professional) · **Russian** — B2 (professional working)